

CONTROLLED DUAL-ATTENTION APPROXIMATION REVEALS DATASET-DEPENDENT WINDOW SENSITIVITY IN MEDICAL IMAGE SEGMENTATION

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ABSTRACT

Dual attention is effective for medical image segmentation, but its quadratic cost has motivated windowed, low-rank, and grouped approximations whose effect on accuracy across tasks is rarely examined. We present ApproxDA-TransUNet, which replaces the dual-attention blocks of DA-TransUNet with an approximation controlled by window size M , projection rank r , and channel group number G . On Synapse multi-organ CT, Kvasir-SEG, and ISIC 2018, we observe strongly dataset-dependent window sensitivity: the Dice range across window sizes is 2.30 points on Synapse but only 0.64 and 0.50 points on Kvasir-SEG and ISIC 2018. With an appropriately selected window, ApproxDA-TransUNet improves Dice over DA-TransUNet on all three datasets under matched evaluation. Attention approximation is therefore not accuracy-neutral, and its window scale is a task-dependent design choice.

Index Terms— Medical image segmentation, dual attention, attention approximation, window size, ablation study

1. INTRODUCTION

Convolutional encoder-decoders such as U-Net [1] and its variants [2, 3, 4, 5] remain strong baselines for medical image segmentation but have limited receptive fields. Transformer-based models address this with global self-attention, either in hybrid CNN-Transformer designs such as TransUNet [6], UCTransNet [7], TransNorm [8], and MIM [9], or in largely Transformer-based designs such as Swin-Unet [10], UNETR [11], and DAE-Former [12]. Dual attention [13] combines a Position Attention Module (PAM) over spatial locations with a Channel Attention Module (CAM) over channels; DA-TransUNet [14] inserts it into TransUNet, and later works extend it to multi-scale [15] and state-space [16] models.

Full PAM costs $\mathcal{O}(N^2C)$ for N positions and C channels. Windowed attention [17], low-rank projection [18], and grouped channel interaction [19] reduce this cost and are usually treated as accuracy-neutral. Whether they change segmentation accuracy, and whether the effect is the same across tasks, has received little controlled study.

We address this question with ApproxDA-TransUNet, which replaces the dual-attention blocks of DA-TransUNet

with approximated ones. With the backbone, training, and evaluation fixed, we vary one approximation parameter at a time on three datasets. Our contributions are:

- **A controlled dual-attention approximation framework.** Window size M , rank r , and group number G control the spatial scope, the affinity resolution, and the channel interaction scope of dual attention, respectively, so that each factor can be ablated in isolation.
- **Evidence of dataset-dependent window sensitivity.** Synapse multi-organ CT is substantially more sensitive to M than Kvasir-SEG and ISIC 2018, and with an appropriately selected window ApproxDA-TransUNet improves Dice over DA-TransUNet on all three datasets under matched evaluation.

2. METHOD

2.1. Architecture overview

ApproxDA-TransUNet keeps the hybrid CNN-ViT encoder and decoder of DA-TransUNet [14], with an R50-ViT-B/16 backbone (Fig. 1). Three CNN stages produce features at strides 2, 4, and 8, and a 12-layer ViT processes 14×14 patch tokens. ApproxDABlocks are placed at the bottleneck and at the three skip connections (512 channels at 28^2 , 256 at 56^2 , 64 at 112^2). The public DA-TransUNet implementation applies dual attention only at the bottleneck and the 512- and 256-channel skips, because its decoder selects the block by the decoder channel count rather than the skip channel count; we keep that implementation unchanged as the baseline and additionally place a block at the 64-channel skip.

Each ApproxDABlock keeps the layout of the DA-TransUNet block and replaces only its attention modules (Fig. 2). Each branch first reduces the C input channels to $C/16$ with a 3×3 convolution, batch normalization, and ReLU, as in DA-TransUNet; the PAM and CAM branches then apply the approximated attention described below, followed by another 3×3 convolution block. The two branches are fused and projected back to C channels.

Approximation axes. The original dual-attention block computes dense spatial and channel attention over the full feature map. ApproxDA-TransUNet approximates it along three independent axes. *Spatial scope* (window size M) con-

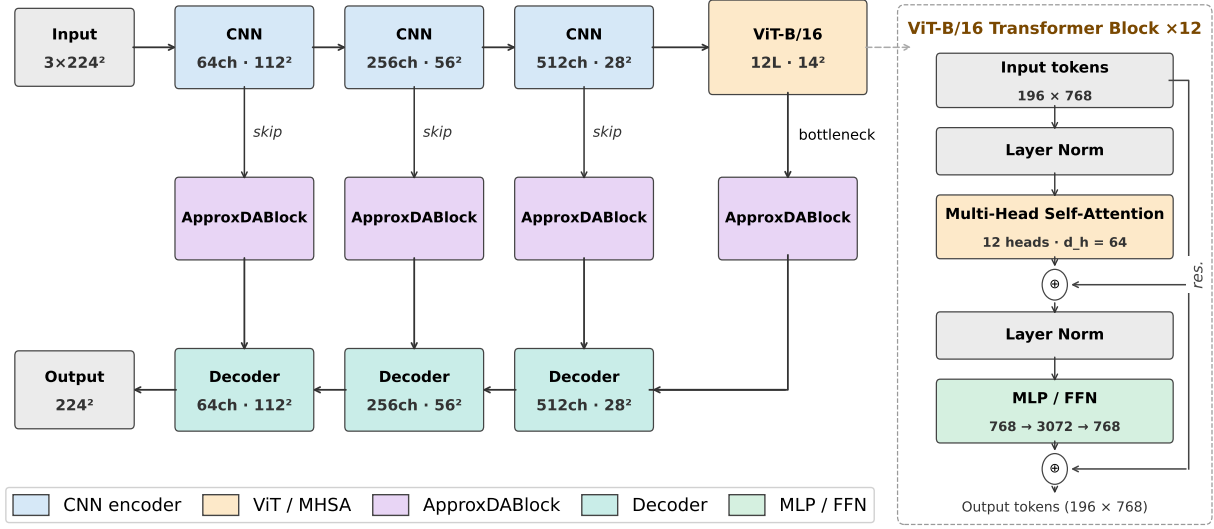


Fig. 1. ApproxDA-TransUNet. ApproxDABlocks (windowed low-rank PAM + grouped CAM + routing) are placed at all three skip connections and at the bottleneck; the CNN–ViT backbone of DA-TransUNet is unchanged.

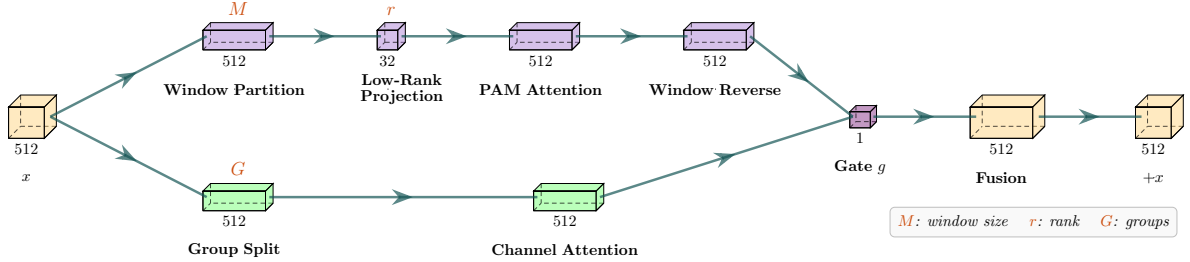


Fig. 2. ApproxDABlock with three independently controllable approximation axes: spatial scope (M), affinity resolution (r), and channel interaction (G). Each branch reduces the input to $C/16$ channels before attention, as in DA-TransUNet; the branch outputs are fused by the per-channel routing vector g and projected back to C channels by a 1×1 convolution.

trols how far PAM reaches; $M \times M$ windows trade long-range context for lower cost. *Affinity resolution* (rank r) controls how finely attention is computed inside a window; higher r approaches exact attention. *Channel interaction* (groups G) restricts CAM to interactions within G channel groups. Any two of $\{M, r, G\}$ can be held fixed while the third varies.

2.2. Spatial approximation

Full PAM. Following [13], let $\mathbf{A} \in \mathbb{R}^{C \times N}$ be the (channel-reduced) input of the attention module, $N = HW$, and let $\mathbf{B}, \mathbf{C} \in \mathbb{R}^{C' \times N}$ and $\mathbf{D} \in \mathbb{R}^{C \times N}$ be query, key, and value features produced by 1×1 convolutions, with columns $\mathbf{B}_j$, $\mathbf{C}_i$, $\mathbf{D}_i$ and $C' = \max(1, C/8)$ as in the DA-TransUNet im-

plementation. PAM computes

$$s_{ji} = \frac{\exp(\mathbf{B}_j^\top \mathbf{C}_i)}{\sum_{i'=1}^N \exp(\mathbf{B}_j^\top \mathbf{C}_{i'})}, \quad \mathbf{E}_j = \alpha \sum_{i=1}^N s_{ji} \mathbf{D}_i + \mathbf{A}_j, \quad (1)$$

where α is a learned scalar initialized to 0. The $N \times N$ attention matrix costs $\mathcal{O}(N^2 C)$.

Window partitioning. Following [17], the map is split into non-overlapping $M \times M$ windows with $M \leftarrow \min(M, H, W)$, and Eq. (1) is evaluated inside each window over its $N_w = M^2$ positions. This reduces the cost to $\mathcal{O}(NM^2 C)$; the window outputs are then merged back to the $H \times W$ grid.

Low-rank projection. Following [18], keys and values inside each window, $\mathbf{C} \in \mathbb{R}^{C' \times N_w}$ and $\mathbf{D} \in \mathbb{R}^{C \times N_w}$, are compressed along the token axis by a learned projection $\mathbf{P} \in$

$\mathbb{R}^{N_w \times r}$ shared across windows:

$$\tilde{\mathbf{C}} = \mathbf{C}\mathbf{P} \in \mathbb{R}^{C' \times r}, \quad \tilde{\mathbf{D}} = \mathbf{D}\mathbf{P} \in \mathbb{R}^{C \times r}. \quad (2)$$

Each query then attends to the r projected keys instead of the N_w original ones:

$$\tilde{s}_{jk} = \frac{\exp(\mathbf{B}_j^\top \tilde{\mathbf{C}}_k)}{\sum_{k'=1}^r \exp(\mathbf{B}_j^\top \tilde{\mathbf{C}}_{k'})}, \quad \mathbf{E}_j = \alpha \sum_{k=1}^r \tilde{s}_{jk} \tilde{\mathbf{D}}_k + \mathbf{A}_j, \quad (3)$$

so the per-window attention matrix is $N_w \times r$ instead of $N_w \times N_w$, and the total cost becomes $\mathcal{O}(NrC)$. As in [13], no temperature scaling is applied. When M is clamped to a smaller map, the first N_w rows of $\mathbf{P}$ are used. Hence $M=112$ gives a single window per map (global spatial scope), but the attention remains rank- r and is *not* exact full PAM; omitting the projection ($\tilde{\mathbf{C}}=\mathbf{C}$, $\tilde{\mathbf{D}}=\mathbf{D}$) recovers exact attention within each window.

2.3. Channel approximation

CAM [13] computes a $C \times C$ channel affinity at $\mathcal{O}(C^2N)$ cost. We split the C channels into G groups of $C_g = C/G$ channels and compute attention within each group. For group g with channel rows $\mathbf{A}_1^{(g)}, \dots, \mathbf{A}_{C_g}^{(g)} \in \mathbb{R}^N$,

$$x_{ij}^{(g)} = \frac{\exp(-\mathbf{A}_i^{(g)} \cdot \mathbf{A}_j^{(g)})}{\sum_{k=1}^{C_g} \exp(-\mathbf{A}_i^{(g)} \cdot \mathbf{A}_k^{(g)})}, \quad (4)$$

$$\mathbf{E}_i^{(g)} = \beta \sum_{j=1}^{C_g} x_{ij}^{(g)} \mathbf{A}_j^{(g)} + \mathbf{A}_i^{(g)},$$

where β is a learned scalar initialized to 0. Following the official DANet and DA-TransUNet implementations, the channel affinity is negated before the softmax (implemented as $\max_k \mathbf{A}_i \cdot \mathbf{A}_k - \mathbf{A}_i \cdot \mathbf{A}_j$, which is equivalent under the softmax). Grouping reduces the cost to $\mathcal{O}(C^2N/G)$, and $G=1$ recovers the CAM module of DA-TransUNet exactly. When a reduced map has fewer than G channels (4 channels at the 64-channel skip), one channel per group is used.

2.4. Routing and complexity

Let $\mathbf{X}$ be the block input with C channels, and let ϕ denote a 3×3 convolution with batch normalization and ReLU. The block output is

$$\mathbf{P} = \phi_{51}(\text{PAM}(\phi_{5a}(\mathbf{X}))), \quad \mathbf{Q} = \phi_{52}(\text{CAM}(\phi_{5c}(\mathbf{X}))),$$

$$\mathbf{Y} = \text{ReLU}(\mathbf{W}_f(\mathbf{g} \odot \mathbf{P} + (\mathbf{1} - \mathbf{g}) \odot \mathbf{Q})), \quad (5)$$

where ϕ_{5a}, ϕ_{5c} map $C \rightarrow C/16$ channels, $\mathbf{W}_f$ is a 1×1 convolution back to C channels (with dropout during training), and $\mathbf{g} \in [0, 1]^{C/16}$ is a per-channel routing vector broadcast over positions, set by the routing mode:

PAM-only:	$\mathbf{g} = \mathbf{1}$
CAM-only:	$\mathbf{g} = \mathbf{0}$
fixed:	$\mathbf{g} = 0.5 \cdot \mathbf{1}$
learned:	$\mathbf{g} = \sigma(\mathbf{W}_g \text{GAP}(\mathbf{X}))$

PAM-only and CAM-only routing build only the branch they use. With exact attention (M covering the whole map, no projection), $G=1$, and $\mathbf{g}=0.5 \cdot \mathbf{1}$, the block reduces to the DA-TransUNet block up to a rescaling of $\mathbf{W}_f$. The network is trained with $\mathcal{L} = \frac{1}{2}\mathcal{L}_{\text{Dice}} + \frac{1}{2}\mathcal{L}_{\text{CE}}$, as in DA-TransUNet.

Windowing and low-rank projection reduce the PAM attention matrix from $N \times N$ to $N \times r$, a factor of $N/r \approx 390$ at the 112^2 skip with $r=32$. End-to-end cost is dominated by the unchanged backbone: the ViT encoder alone holds 85.1M of the parameters and 17.4 of the GFLOPs (fvcore, one 224×224 slice), and the CNN encoder and decoder most of the rest. The attention blocks of DA-TransUNet account for 0.87 GFLOPs; those of ApproxDA-TransUNet (PAM-only, $M=28$, $r=32$) for 0.40 GFLOPs, compared with 1.37 GFLOPs for the same blocks with exact attention. The full networks have 107.95M parameters / 30.20 GFLOPs (DA-TransUNet) and 105.99M / 29.73 GFLOPs (ApproxDA-TransUNet); we therefore do not claim substantial whole-network acceleration. **[TODO: optional: block-level latency / memory on the same GPU.]**

3. EXPERIMENTS

3.1. Datasets and protocol

Datasets. *Synapse* [20]: 30 abdominal CT scans, 8 organs, standard 18/12 train/test split. *Kvasir-SEG* [21]: 1000 colonoscopy images with binary polyp masks, using the 880/120 train/test split recommended with the dataset. *ISIC 2018* [22]: dermoscopy images with binary lesion masks, using the official Task 1 training (2594), validation (100), and test (1000) sets.

Model selection. No checkpoint selection is performed: every model, including the re-run DA-TransUNet, is evaluated at its final training epoch, as in TransUNet [6]. On ISIC 2018, M , r , G , and routing are selected on the official validation set. Synapse and Kvasir-SEG have no validation split; following common practice on Synapse [6, 14], their configuration is taken from the ablations in Tables 2 and 3, which report every configuration on the test set (see Limitations).

Training. All models (including the re-run DA-TransUNet) are trained for 300 epochs with ImageNet-21K pre-trained R50-ViT-B/16 weights, 224×224 input, random rotation and flipping, batch size 24, polynomial learning-rate decay, and loss $\frac{1}{2}\mathcal{L}_{\text{CE}} + \frac{1}{2}\mathcal{L}_{\text{Dice}}$, on the same hardware (**one NVIDIA Tesla T4**). Following DA-TransUNet [14], Synapse uses SGD (learning rate 0.01, momentum 0.9, weight decay 10^{-4}) and Kvasir-SEG and ISIC 2018 use Adam (learning rate 10^{-3} , weight decay 10^{-4}). We train longer than the orig-

Table 1. Segmentation results. Rows marked $\dagger$ are values reported in the literature under their original protocols and are shown for reference only; the matched comparison is between the last two rows.

Method	Synapse		Kvasir-SEG		ISIC 2018	
	DSC $\uparrow$	HD95 $\downarrow$	DSC $\uparrow$	mIoU $\uparrow$	DSC $\uparrow$	mIoU $\uparrow$
U-Net $\dagger$ [1]	76.85	39.70	88.22	80.12	87.22	81.14
U-Net++ $\dagger$ [2]	76.91	36.93	86.57	77.67	87.30	81.33
Res-UNet $\dagger$ [4]	76.95	38.44	77.85	66.04	83.32	76.51
MultiResUNet $\dagger$ [5]	77.42	36.84	—	—	—	—
Att-UNet $\dagger$ [3]	77.77	36.02	86.61	78.01	87.60	81.51
TransUNet $\dagger$ [6]	77.48	31.69	87.91	80.03	88.78	82.63
UCTransNet $\dagger$ [7]	78.23	26.75	—	—	—	—
TransNorm $\dagger$ [8]	78.40	30.25	—	—	—	—
MIM $\dagger$ [9]	78.59	26.59	—	—	—	—
Swin-UNet $\dagger$ [10]	79.13	21.55	—	—	—	—
DA-TransUNet $\dagger$ [14]	79.80	23.48	88.47	—	88.88	82.78
DA-TransUNet (re-run)	—	—	88.44	81.70	88.43	80.68
ApproxDA-TransUNet	80.94	27.49	90.17	84.27	89.55	—

ApproxDA-TransUNet: PAM-only routing, $r=32$; M = Synapse 28, Kvasir-SEG 56, ISIC 2018 28, selected from the ablations (Synapse, Kvasir-SEG) or on the validation set (ISIC 2018).

inal protocols (150 epochs on Synapse; 500 and 50 epochs on Kvasir-SEG and ISIC 2018) and use 224×224 instead of 256×256 input on all datasets, so that the same window sizes M apply everywhere. Unless stated otherwise, ApproxDA-TransUNet uses $r=32$ and PAM-only routing; G ($= 8$ by default) only matters for routings that use CAM.

Metrics. Synapse: mean Dice (DSC) and 95% Hausdorff distance (HD95, mm) over the 8 organs. Kvasir-SEG and ISIC 2018: DSC and mean IoU (mIoU); we do not report HD95 in mm for these datasets because pixel spacing is not calibrated. [TODO: main comparison: mean $\pm$ std over 3 seeds.]

3.2. Main results

Under matched evaluation, ApproxDA-TransUNet improves DSC over the re-run DA-TransUNet on **all three** datasets (Table 1, Fig. 3; **+1.73** Kvasir-SEG, **+1.12** ISIC 2018, [TODO: Synapse after re-run]). Gains are metric-specific: Synapse HD95 is **worse** than the reported DA-TransUNet and Swin-UNet values, and ISIC 2018 mIoU is **below** the reported DA-TransUNet value. [TODO: per-organ Synapse DSC; paired test / bootstrap CI (experiments/statistical_validation.py).]

3.3. Window sensitivity

Table 2 varies only M . On Synapse, DSC peaks at an intermediate window ($M=28$) and drops on both sides (range 2.30); Kvasir-SEG and ISIC 2018 are **3.6–4.6 $\times$** less sensitive (0.64 and 0.50). We call this range the *empirical window sensitivity*; it describes a model’s response to M on a dataset, not an independent predictor of contextual requirement. The

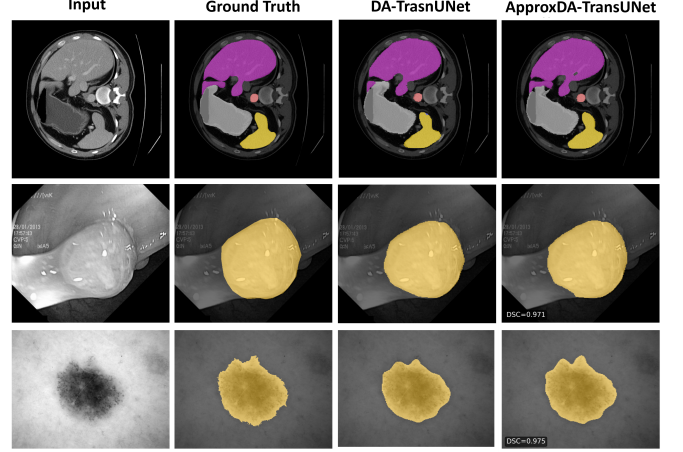


Fig. 3. Qualitative comparison (columns: input, ground truth, DA-TransUNet, ApproxDA-TransUNet). Top: Synapse ($M=28$); middle: Kvasir-SEG ($M=56$); bottom: ISIC 2018 ($M=28$).

Table 2. DSC (%) versus window size M (PAM-only routing, $r=32$). Range = $\max_M \text{DSC}(M) - \min_M \text{DSC}(M)$.

Setting	Synapse	Kvasir-SEG	ISIC 2018
$M=7$	78.64	89.53	89.41
$M=28$	80.94	89.54	89.55
$M=56$	78.90	90.17	89.05
$M=112$ (global, low-rank)	79.44	89.56	89.51
Range	2.30	0.64	0.50
DA-TransUNet (full DA, re-run)	—	88.44	88.43

difference is consistent with multi-organ CT depending more on inter-organ context than polyp and lesion segmentation, which depend mainly on local appearance. The sensitivity also bounds how much the choice of M matters in practice: on Kvasir-SEG and ISIC 2018 any tested window stays within **0.64** points of the best, whereas on Synapse the window must be tuned. On all three datasets the best window **exceeds** DA-TransUNet, suggesting a locality-induced bias. This comparison changes both the attention approximation and the routing (PAM-only instead of PAM + CAM); the exact-PAM row of Table 3 isolates the approximation itself. $M=112$ keeps the rank-32 projection and thus differs from full PAM.

Fig. 4 visualizes this bias on Kvasir-SEG. On 10 test images, windowed PAM ($M=7$) places 62.3% of its contribution inside the polyp region, compared with 34.2% for global PAM ($M=112$), and the windowed share is higher in 9 of the 10 images. Windowed attention concentrates on polyp boundaries, whereas global attention spreads over surrounding tissue. This observation is consistent with the locality interpretation but does not by itself establish a causal link to accuracy.

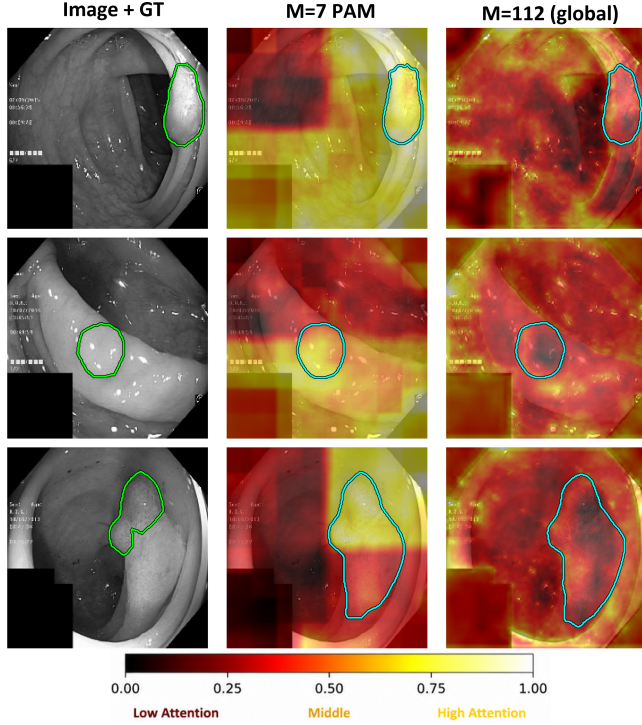


Fig. 4. PAM contribution maps ($\|output - input\|_2$) on three Kvasir-SEG images. Each row: input with ground-truth contour (left), windowed PAM with $M=7$ (center), global PAM with $M=112$ (right). Each panel is normalized independently, so magnitudes are not comparable across panels.

3.4. Rank, grouping, and routing

In Table 3, raising r from 32 to 64 does **not** improve DSC, so window scale matters more than rank in the tested range; the no-projection row is exact global PAM (PAM-only routing), isolating the effect of the approximation from that of the routing, and the window-only row removes the projection at the best Synapse window. G is ablated under CAM-only routing, since CAM does not contribute under PAM-only routing. [TODO: grouping result.] The learned gate converged to $g \approx 0.5$ in our Synapse experiments and did not outperform PAM-only routing, which we therefore use elsewhere.

Limitations. Because Synapse and Kvasir-SEG have no validation split, their configurations were selected from test-set ablations, so the reported ApproxDA-TransUNet results on these datasets may be optimistic. The bias is bounded by the window-sensitivity range: small on Kvasir-SEG (0.64 points), but up to 2.30 points on Synapse, where fine window selection is precisely what our results call for; in practice, M should be tuned on held-out data. We also use three datasets and one backbone, and whether window sensitivity can be predicted without window ablations remains open.

Table 3. Single-factor ablations on Synapse. Each block varies one factor; the others are fixed as indicated.

Factor (fixed)	Setting	DSC $\uparrow$	HD95 $\downarrow$
Rank r ($M=112$, PAM)	$r=16$	—	—
	$r=32$	79.44	27.26
	$r=64$	78.93	31.21
	no projection	—	—
Window only ($M=28$, PAM)	no projection	—	—
Groups G (CAM-only)	$G=1$	—	—
	$G=4$	—	—
	$G=8$	78.26	30.59
Routing ($M=7$, $r=32$)	PAM-only	78.64	31.09
	CAM-only	78.26	30.59
	fixed $g=0.5$	—	—
	learned	77.78	34.29

4. CONCLUSION

Controlled approximation of dual attention shows that sensitivity to the window size is strongly dataset-dependent: large on multi-organ CT, small on polyp and skin-lesion segmentation. With an appropriate window, ApproxDA-TransUNet improves Dice over DA-TransUNet on all three datasets. Attention approximation should be treated as a task-dependent design choice rather than an accuracy-neutral efficiency measure.

5. COMPLIANCE WITH ETHICAL STANDARDS

This study was conducted retrospectively using human subject data made available in open access by the Synapse [20], Kvasir-SEG [21], and ISIC 2018 [22] challenges. Ethical approval was not required as confirmed by the licenses attached with the open access data.

6. ACKNOWLEDGMENTS

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